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Learning Effect is Demonstrated in Vision-Altering Prism Goggles

Abstract

Conditions such as right-hemisphere stroke lead to spatial neglect and a reduced ability to complete visuomotor tasks. Prism lens therapy has been shown to be effective as a treatment for visual conditions such as strabismus, as well as efficacy in post-stroke therapy. Here we examined the learning effect present in completion of a visuomotor task while wearing vision-altering prism goggles to better understand the way in which the brain can change its forward model and adapt to altered vision. The hypothesis that the time it takes to complete a simple visuomotor task will decrease the longer the subject wears the vision-altering prism goggles was tested. 10 subjects were given a visuomotor task while wearing one pair of goggles that skew 20° to the left and subsequently one pair that skew 20° to the right, as well as pre- and post-trial controls without goggles. Results supported this hypothesis and demonstrated a potential motor learning effect in addition to the visual learning effect.

Introduction

In this experiment we sought to answer the question of what kind of learning effect(s), if any, are occurring when vision is altered. We examined this question by using vision-altering prism goggles, which have previously been shown to be a reliable method of precisely altering vision for experimentation.[1,2] This experiment and others like it are important, as prism lens therapy has been shown to help those with post-stroke spatial neglect regain a degree of normal function in the neglected space, improving important cognitive skills such as hand-eye coordination.[3] Understanding the learning effects present in visuomotor tasks when vision is altered can help us better understand the way diseases that affect vision and coordination operate and develop improved therapeutic methods for treatment of these diseases.

By giving each subject a timed visuomotor task before, during, and after wearing vision-altering prism goggles, we were able to see whether a learning effect is in place, and

whether an aftereffect is present once the goggles are removed. In this experiment, we tested the hypothesis that the time to place the block in the target area will decrease between the first and last trials of each set of prism goggles due to the learning effect.

Methods

Measuring Prism Goggle Skewness

Before experimental trials were conducted, the skewness of the prism goggles was measured to ensure consistent results. Two sets of prism goggles were used for the duration of the experiment, one left-skewed pair, referred to as O3, and one right-skewed pair, referred to as Y3. A red laser pointer was set firmly on top of an elevated box and aimed at a fixed distance from a blank wall. The point at which the laser hit the wall was marked with a pencil. Each set of goggles were then held level with the angle of the laser and perpendicular to its beam directly in front of the laser, and the new point at which the beam fell on the wall was marked with a pencil. The distance from the goggles to the wall and the distance between the two points on the wall was then measured using a tape measure, representing the adjacent and opposite sides of a triangle respectively (Fig 1). The angle θ between the hypotenuse and adjacent sides was then calculated using the formula $\theta = \arctan(\text{opposite}/\text{adjacent})$. This angle θ was the degree of skewness. In both goggles we found θ to be approximately 20°.

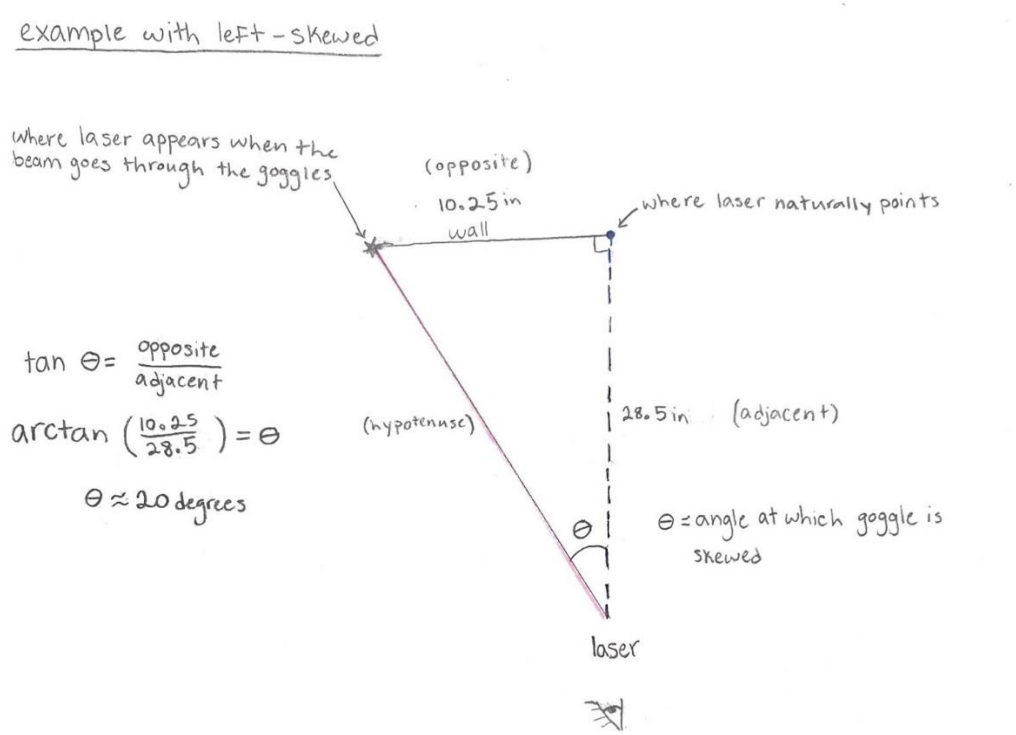

Figure 1 Schematic representation of how the angle of skewness, θ, was calculated using left-skewed goggles. The laser was pointed at the wall through the goggles when stationary and the opposite and adjacent sides measured. Arctan was then used to calculate θ.

Experimental Procedure

10 subjects were used in this experiment. Each subject was seated in front and directly in the middle of a sheet of cardboard placed on a table 1 inch from the edge at approximately elbow height and a 150g cardboard block was placed in front of them on the cardboard (Fig 2). The cardboard was divided into 3 equal sections, one left, one middle, and one right. Across the top of the cardboard were 3 differently colored squares made of colored tape and the same dimensions of the block, one for each section. These served as our control areas, and five Xs made with colored tape served as target areas. On the reverse side of the cardboard was another six target areas created as boxes using colored tape (Fig 3). Before any trials began, the subject was given instructions on the proceeding trials and the names of each color established. The subject was first given three control trials in which the color conductor, stood on the other side of the table, called out one of the three colors of control areas, and the subject instructed to place the block on the area immediately but without

rushing until all three control areas were called once. The time from the color being spoken to the block being placed was then recorded and logged. After all three control areas were called out, the subject was fitted with the O3 left-skewed goggles and instructed to perform the same task with the target areas, with each trial area being randomly selected by the color conductor. This was repeated until 10 trials were completed, then the subject was given the Y3 right-skewed goggles, and the cardboard flipped to the other side to be repeated for another 10 trials. After this, the goggles were removed, and the cardboard flipped back over to the first side. The subject was then instructed to complete the three control trials just as before, for a total of 3 pre-trial controls, 20 trials, and 3 post-trial controls.

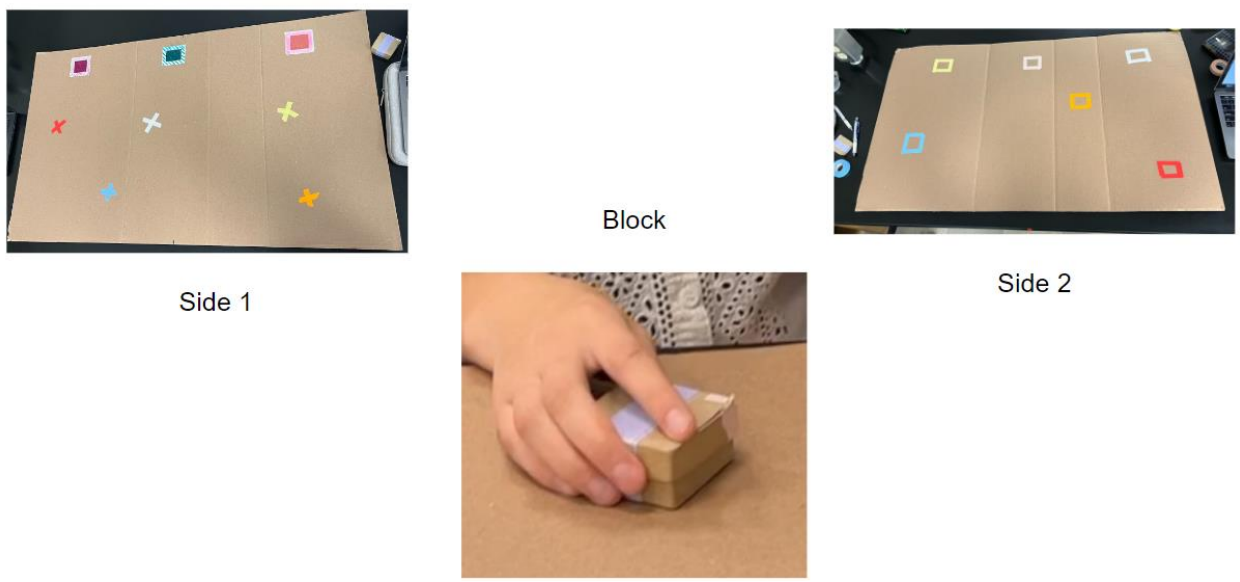

Figure 2 Pictures of the target areas and block. On the left, titled "Side 1", are the three control areas and five target areas. On the right, titled "Side 2", are the remaining target areas. In the middle, titled "Block", is the block placed on the target areas.

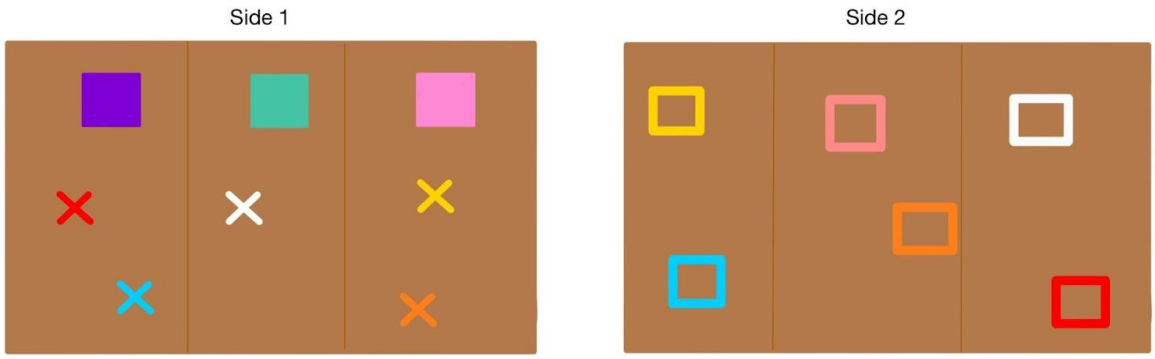

Figure 3 Schematic representation of the target areas. On the left, titled "Side 1", are the three control areas and five target areas depicted by the colored boxes and Xs. On the right, titled "Side 2", are the remaining target areas depicted by the colored boxes.

Results

Left-Skewed and Right-Skewed Goggles Showed Decrease in Trial Time

The recorded time per trial of each subject was analyzed to see the relationship between the length of time goggles were worn and the speed to place the block on the trial area after the color was read out. Time was measured in seconds, and the trial number was used as an indication of the length of time the goggles were worn by the subject. These data were first plotted as a line scatter plot for each pair of goggles to qualitatively examine each subject's trials from start to finish, as well as between each other, to ensure no significant outliers (Fig. 4). Results of the averaged trial times indicate a linear decrease in time to complete each trial per subject with each subsequent trial in both the left-skewed and right-skewed goggles, with a significant decrease being seen when comparing the first and last trials of each pair of goggles (Fig. 5). On average, left-skewed goggles showed a 45.2% increase in the speed to complete trials from start to finish, while right-skewed goggles showed a 60.1% increase. This difference between each pair of goggles may be due to the qualitative difference in the right-skewed film applied to the goggles, as subjects frequently reported the right-skewed goggles feeling more disorienting despite having the same angle of skew as the left-skewed goggles.

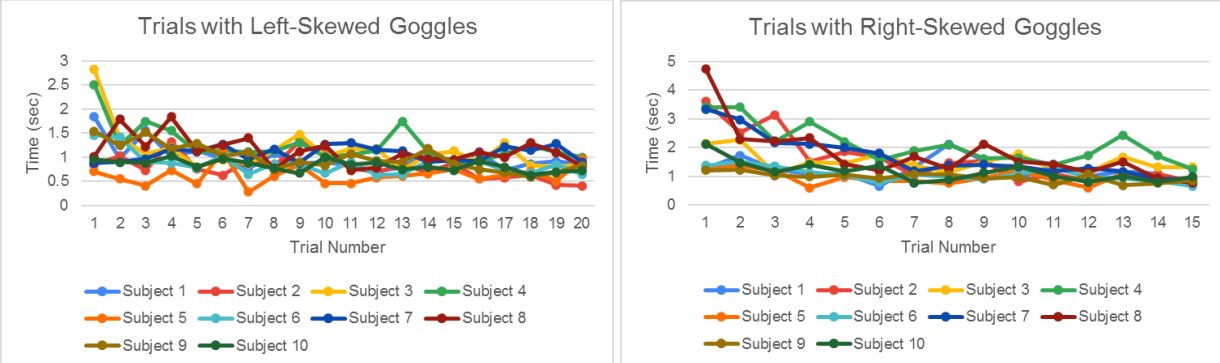

Figure 4 A) Combined scatter plots of trials per subject with left-skewed goggles for all subjects on the left and right-skewed goggles on the right. The subsequent trial numbers are represented on the x-xaxis. Time, measured in seconds, is on the y-axis. Each colored line represents a single subject as denoted in the legend.

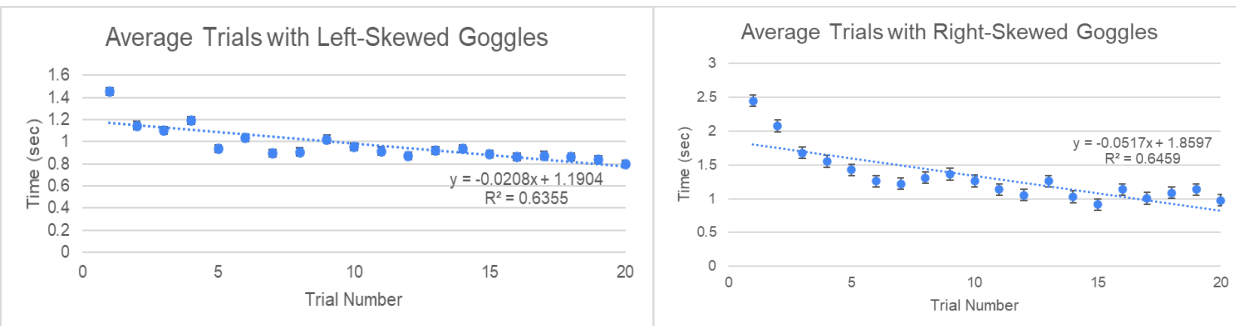

Figure 5 Combined average of all trials with left-skewed goggles on the left and right-skewed goggles on the right, depicting the linear relationship of trial number and trial time. The subsequent trial numbers are represented on the x-xaxis. Time, measured in seconds, is on the y-axis. Standard error bars are present but barely visible due to small values. The dotted line in each scatter plot represents the line of best fit.

Normalized Trial Times Account for Subject Differences

These data were further analyzed by normalizing the averaged dataset of both the left-skewed and right-skewed goggles separately. This was done to eliminate any outliers while also allowing a better comparison to be drawn between subjects, as this accounts for differences between subjects, such as those arising from differences in average reaction time and physiology. These normalized data were then plotted as histograms to analyze the standard error and averaged reaction times of each subject with each other (Fig. 6).

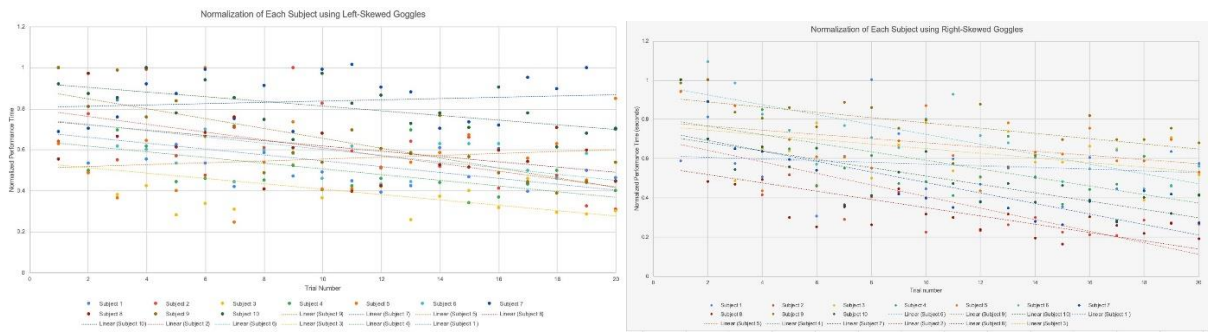

Figure 6 Normalized average trial time of each subject for each pair of goggles. Left: Normalized average time per subject across all trials with left-skewed goggles. Right: Normalized average trial time per subject across all trials with right-skewed goggles.

Pre-Trial & Post-Trial Controls Showed Decrease in Time

The pre-trial controls and post-trial controls were averaged across all subjects and then compared with each other for each of the three control trials (Fig. 7). Results showed a significant decrease in average trial time across all subjects when comparing the pre-control and post-control trials for each of the three control trials. If visual adaptation were the only learning effect affecting results, we would expect the bars to be relatively the same within each control area. This decrease thus suggests other learning effects may be affecting performance, such as motor learning, rather than just visual adaptation.

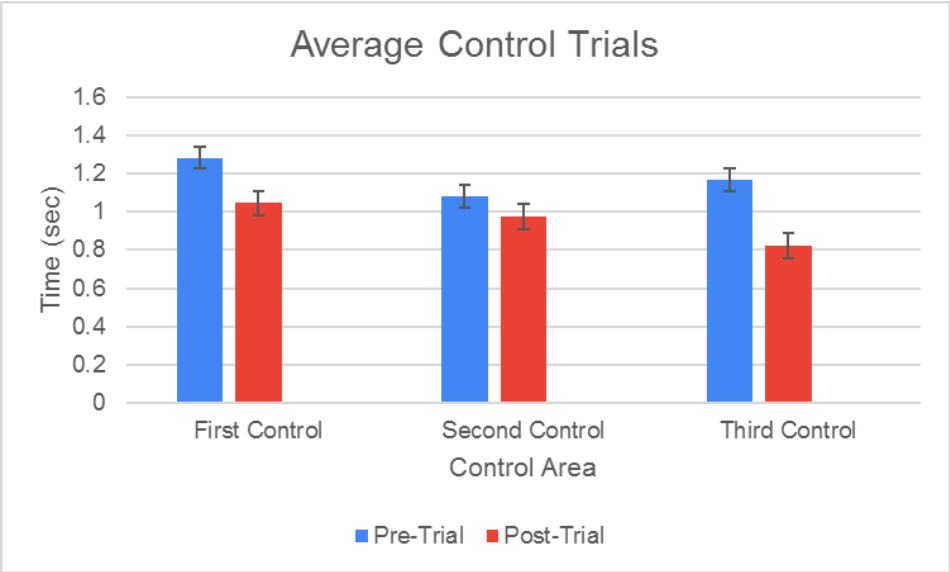

Figure 7 Histogram comparing average pre- and post-trial controls. Post-trials in red show significant decrease in average trial time compared to pre-trial controls depicted in blue, suggesting multiple types of learning effects.

Discussion

In summary, our hypothesis was supported. A significant decrease in trial time was observed between the first and last trials for each pair of left-skewed and right-skewed prism goggles. In addition, the results of the pre-trial and post-trial controls demonstrate a potential motor learning effect alongside visual adaptation affect the ability to adapt to the prism goggles when performing the visuomotor task.

Potential sources of error include the length of time it took for subjects to complete the trials, as those who took longer to complete trials had a larger amount of time to adapt. The noise in the lab room was also not controlled for, and those who completed trials when the room was noisier may have had a more difficult time interpreting and reacting to the instructions. Differences in the size of subjects may have contributed to error, as those

shorter and closer to the board would have had a more difficult time finding target areas in their periphery. Some subjects were more competitive than others and were thus responding much quicker and more vigorously than other subjects. Finally, some subjects also confused colors, such as red and pink, in the middle of trials, and these reaction time differences may have contributed to the difference in time. These sources of error could be fixed by standardizing the allowed time to complete each trial, standardizing the head position by raising or lowering the chair so the subjects' heads are always at the same height, and by performing the trials in a quiet room.

In 1999 Juan Fernández-Ruiz and Rosalinda Díaz demonstrated that the process of adaptation is dependent on the number of interactions between the visual and motor system and not on the time spent wearing prism goggles.[1] This is an interesting result with respect to our hypothesis, which is contingent on the number of trials as a marker of adaptation. Our results remain consistent with those found by Fernández-Ruiz and Díaz, but further experiments would be necessary to distinguish whether the adaptation observed in both experiments are dependent on the interactions between the visual and motor system or are merely dependent on the length of time with the prism goggles on. For example, repeating trials while counting the amount of discrete motor actions, such as picking up the block, moving the block, and placing the block, while using different visual cues such as color cards, would allow for the relationship between the visual and motor system to be better distinguished. They also demonstrated that the magnitude of the aftereffect is highly correlated with the magnitude of the adaptation, which suggests that the aftereffect does not directly mirror the adaptation process and thus there are yet unknown factors that affect each process independently.

In a similar experiment from 2005, Redding et al. demonstrated a visuomotor adaptation exists when given a prism goggle task by showing how the brain adapts to changes in visually perceived coordinates of objects in space and showing that the brain generates a new forward model from prediction errors.[2] This paper's findings are consistent with our results as well, and provide an explanation of how the brain is adapting to the continued effects of wearing prism goggles. Repeating our experiment with a larger amount of post-trial controls would allow us to see whether their results demonstrating a

negative aftereffect, in which errors in block placement are now present in the opposite direction, are also present in our experimental design and better understand the brain's method of generating new forward models.

Future directions to continue this research could include using larger and smaller angles of skew in the goggles to examine the rate of adaptation. For example, one could examine the hypothesis that 40° vision-skewing prism goggles will lead to a slower rate of adaptation than 20° vision-skewing prism goggles of the same direction of skew. Understanding this relationship between rate of adaptation and the angle of skew could provide more precise treatment for a variety of conditions such as strabismus. One could also examine the difference in adaptation when affecting only one eye compared to both eyes by putting only a single vision-skewing lens on the left and right eyes separately and comparing them to trials where both eyes are affected. Understanding how monocular vision changes compared to binocular vision has implications in better understanding the way in which right-hemisphere stroke affects vision and develop improved post-stroke therapies.

References

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